

Lead Scoring Assignment



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Course:
AI & Data Science

Tools Used:
Python , Pandas, Scikit-Learn, Jupyter Notebook

Problem Statement



- The company wants to identify potential leads that are most likely to convert into customers.
- A lead scoring model is developed using Logistic Regression.
- The model helps the sales team focus on high-quality leads.
- This improves marketing efficiency and reduces unnecessary effort.

Objective of the Project

- Build a Logistic Regression model for lead scoring.
- Predict whether a lead will convert into a customer.
- Identify important variables affecting lead conversion.
- Help the sales team focus on potential customers.
- Improve overall marketing and business efficiency.

Dataset Overview

- Dataset Name : Leads Dataset
- Total records : 9240
- Number of features : 35
- Target Variable: Converted
- Dataset contains customer interaction and lead information.

Data Preprocessing

- Missing values were identified and handled.
- Unnecessary columns were removed from the dataset.
- Categorical variables were converted into numerical format.
- Feature scaling was performed using StandardScaler.
- Dataset was divided into training and testing data.

Model Building

- Logistic Regression algorithm was used for model training.
- The model was trained using X_train and Y_train datasets.
- Predictions were generated using the test dataset(X_test).
- Accuracy score, classification report, and confusion matrix were used for evaluation.
- Class balancing was applied using class_weight = 'balanced'.

Model Performance

- Model Accuracy achieved was approximately 45%.
- Classification report was generated for performance evaluation.
- Confusion matrix was used to analyze prediction results.
- The model successfully identified potential lead conversion patterns.
- Accuracy can be improved further using advanced feature engineering and tuning.

Key Insights

- Total Time Spent on Website was one of the strongest predictors of lead conversion.
- Total Visits showed significant impact on customer interest.
- Lead Source helped identify high-quality leads.
- Highly engaged customers had higher conversion probability.
- Lead scoring can help prioritize potential customers effectively.

Business Recommendations

- Focus marketing efforts on leads with higher website engagement.
- Prioritize leads with higher visit frequency and interaction levels.
- Improve targeting strategies based on lead source analysis.
- Use lead scoring to reduce unnecessary sales efforts.
- Continuously monitor and improve model performance using updated data.

Conclusion

- Logistic Regression was successfully implemented for lead scoring analysis.
- The model helped identify important factors affecting lead conversion.
- Lead scoring can support better decision-making for the sales team.
- The project demonstrated the complete machine learning workflow from preprocessing to evaluation.
- Future improvements can enhance prediction accuracy and business impact.

Thank You